

Exact or Nothing

Gate-Governed Measurement of the Liquidation Map on a Transparent Perpetual-Futures Venue

nat2_exact_map_preprint.tex · 11pp · Aug 2026 · code: ~/nat2 (separate repo) · numbers: nat2/data/ledger.jsonl

Thesis

Cascade research is normally done on an *estimated* liquidation map (aggregate OI $\times$ assumed leverage mix). On Hyperliquid it need not be: per-position `liquidationPx` is published, mark comes from a published oracle, any address's fills are public. Refuse estimation entirely and see what the programme looks like — and what becomes visible that estimation hid.

The two rules (enforced in code, not prose)

- **Exact or nothing.** An unobserved position is *absent* and enters the coverage deficit; it is never imputed. Asymmetry: an 80%-right approximation gives a confidently wrong backtest; an acknowledged hole gives a coverage number, and a number can gate a decision.
- **Gates before models.** `feed` $\rightarrow$ `map` $\rightarrow$ `magnet` $\rightarrow$ `persistence` $\rightarrow$ `decay`. Each may end its branch; two may end the project. The refusal is the product.

Substrate

Two clocks. Every record carries t^{event} (venue) and t^{ingest} (first knowable). Causal readable set $\mathcal{F}_T = \{r : t^{\text{ingest}}(r) \leq T\}$; a replay audit fails on any diff. Lookahead becomes *queryable*, not a code-review hope.

WORM store. Hourly rotation, sha256 into a manifest on close, never reopened. Restart mid-hour $\Rightarrow$ new part, so a changed digest always means corruption, never a bounced daemon. Per-stream `seq` survives restarts $\Rightarrow$ a hole means records *lost*.

Ledger. $h_i = \text{SHA256}(\text{enc}(i, \tau_i, \kappa_i, \pi_i, h_{i-1}))$, $h_{-1} = 0^{64}$. Threat model is not an attacker — it is the researcher. The 14 variants tried before the one that worked stay on the record.

Admissibility. Run iff $V(g)$ exists $\wedge$ $\text{pass}(g) = 1 \wedge t - \tau(g) \leq \Delta$ ($\Delta = 24h$). Without the freshness clause a gate that passed once becomes a permanent licence.

Coverage, and the factor of two

$$\hat{\rho}_t = \frac{N_t}{s \cdot \text{OI}_t}, \quad s \in \{1, 2\}$$

OI counts each contract once, but a wallet registry sees both sides $\Rightarrow$ outstanding notional is 2 OI. $s=2$ is conservative, is printed on every map, and is an *open verify item*: being wrong here is a factor of two on the one number the map is judged by.

Measured (ledger, with dates)

gate feed	PASS, 20 checks, 0 failed
gate map	FAIL on predictive
Coverage $s=2$	BTC .315 / ETH .380 / SOL .247 (floor .25)
Derivation exact ($< 10^{-4}$)	40.8% of 2,926 positions
median / p90 rel err	0.152 / 0.921
no-isolated-margin	58% of 1,074 cross positions
Registry	3,596 wallets; sweep ≈ 5 min = whole 1,200/min IP budget
Leaderboard probe 08-07	41,392 wallets; equity-2k = 61% OI / 28% active; volume-2k = 39% / 46%; union 69%

[!] The 69% probe is *not* the coverage number. Its ratio to the ledger's 25–38% is $\approx s$ itself — plausibly one-sided, unresolved. Report both, never the flattering one.

The finding (negative, and general)

Of 48 observed liquidations the equity-seeded registry held

2.9% of liquidated *wallets*, **21.9%** of liquidated *notional*

($\$223,491$, 35 wallets; re-read next cycle 2.8% / 21.9% over 49). Mass and events are carried by *different populations*: the map wants whoever holds size, but whoever gets liquidated is small and levered. So the OI rescaling of the heatmap paper cannot repair it — that correction assumes uniform thinning, and this is biased in *shape*. Generalises to any venue where positions are sampled rather than enumerated: a single notional-denominated coverage scalar reports the flattering half.

Why the headline is not believable yet

Mapped notional fraction reads 0.0% — and that is *confounded, not informative*: a liquidated wallet no longer holds the position, so a snapshot taken after the event has nothing to have predicted. Cannot yet separate “different populations” from “we looked too late”.

The discriminating experiment (pre-stated in the ledger)

`nat2 cycle --snapshot-every 6h --scan-every 1h`, one overlap reading appended per pass; needs $\sim$ a day.

1. mapped frac climbs off 0 $\Rightarrow$ per-position scoring works, instrument stands.
2. mapped pinned at 0, notional holds $\approx 22\%$ $\Rightarrow$ we see the right wallets but never *before* they blow up $\Rightarrow$ demote to cluster-level scoring (weaker, still falsifiable).
3. notional frac collapses too $\Rightarrow$ seed a third registry by *liquidation risk* (margin ratio / distance-to-liq), computable from data already held.

Only (1) validates as specified. The discriminating variable is calendar time under capture — hence capture runs while the rest is built; a point-in-time series cannot be recovered later.

Where liquidations are observable

Public tape carries no marker. `userFills` does, twice: the liquidated wallet's own dir string, and — far more abundantly — the *counterparty's* fill, a `liquidation` object naming `liquidatedUser`, `markPx`, method. Measured 08-08 over 117,485 fills / 70 wallets: exactly *one* self-side dir, but 16/70 wallets had taken the other side. Efficient observer set is a handful of high-volume MMs, not the registry. REST has no per-user cap (WS caps at 15). Scoring drops events predating their snapshot and reports the count: 24 of the first 48.

Venue-native quantities with no CEX analogue

$\pi_t = (p^{\text{mark}} - p^{\text{oracle}}) / p^{\text{oracle}}$ replaces a second-venue sidecar (no second clock/cost model) · hourly funding is first-order at cascade horizons · Δ^{HLP} measures unabsorbed cascade flow (exhaustion observed, not inferred) · one address's perp+spot+vault visible $\Rightarrow$ intra-venue hedges detectable.

Open / unverified (treated as live defects)

s convention · liq-formula residual (size-tiered maintenance margin prime suspect) · mark construction & which price triggers · funding clamp/cadence · rate-limit table (stamped, warns at 90d) · WS connections per IP · node-data cost (sizes M3). Gaps: 12book 45 records/70s across 3 coins (sparser than hinted); clock skew flipped $-196\text{ms} \rightarrow +395\text{ms}$, within 2s tolerance but unexplained.

Layers (calls go downward only)

L-1 capture (WORM, dual-stamped, daemon) · L0 features (pure bar $\rightarrow$ vector, causality contract) · L1 experts (calibrated *p* or score) · L2 meta (Hedge / fixed-share over net-of-cost expert PnL) · L3 allocate (vol target $\rightarrow$ capacity cap $\rightarrow$ maxLeverage cap $\rightarrow$ liq-distance floor) · L4 execute (simulator | testnet | mainnet, flag-gated off) · L5 ledger. Downward-only is what lets simulator and live path share code honestly.

Milestones — each ends at a gate

M0 client + capture + WORM $\rightarrow$ **gate feed done** · M1 reconstruction + map + coverage $\rightarrow$ **gate map built, FAIL** · M2 labels + magnet experts + purged WFO $\rightarrow$ **gate magnet** · M3 node backfill + vault tagging $\rightarrow$ **gate persistence** (*go/no-go for smart money*) · M4 cohort flow + decay study $\rightarrow$ **gate decay** · M5 meta + sizer + simulator + testnet · M6 TUI. M3/M4 precede M5 deliberately: each is $\sim$ a week and each may end its branch.

Expert contract

`fit` / `predict` / `baseline()` **mandatory** — GBDT-vs-sign(imb) and clone-vs-raw-flow are the same rule, written once, enforced for every expert by `nat2 eval`. Must clear edge $> 2(\text{fees} + \text{funding} + \text{slippage}) / \sigma$ net, or it does not enter the pool. LightGBM depth 3 (short history punishes variance; value is in features and labels). Purged walk-forward, embargo *h*, isotonic calibration on OOS folds only. Two-barrier race (Stage A), triple barrier on touch (Stage B), uniqueness weights — cascade episodes overlap heavily. Barriers evaluate on *mark*, because that is what the position experiences.

Named risks

Reconstruction drift (continuous sampled reconcile; drift FAILs **gate feed**) · cross/isolated modelled wrong (both diffed vs HL's own `liquidationPx`) · overlapping-cascade overfit · **crowding** — **the exact map is exact for hunters too** (edge is the conditional odds *after* the touch; fixed-share demotes decayed experts; our own liq distance is watched by the same code that builds the map) · oracle-manipulation tail, JELLY-style · short history (wider embargo, smaller models, *never* pad with CEX data) · single-venue operational risk, accepted explicitly.

Siblings

#3 heatmap supplies the model, this supplies the instrument (and revises #3's obstacle) · #2/#7 supply the adverse-selection barrier the execution layer is built around · #6 process is the same gate discipline one level up.